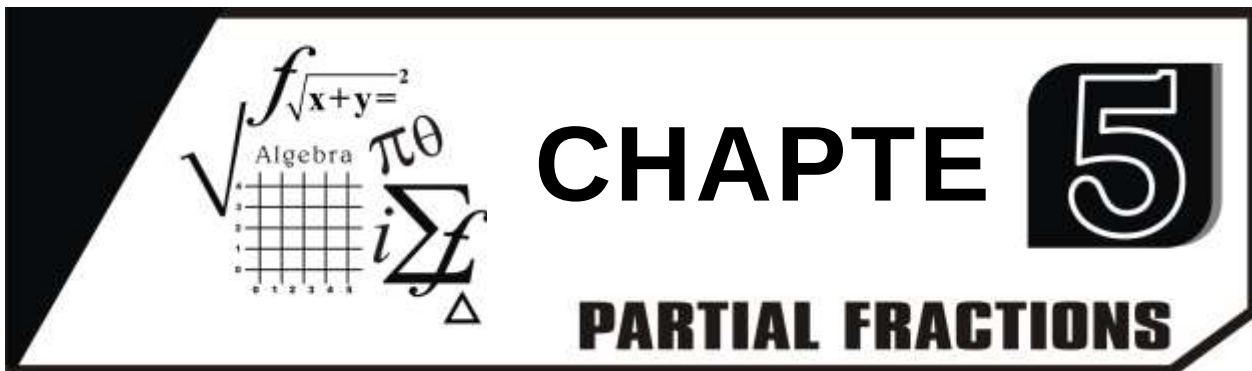




CHAPTE 5

PARTIAL FRACTIONS

Rational Fractions:

A fraction $\frac{P(x)}{Q(x)}$ where both $P(x)$ and $Q(x)$ are polynomials and $Q(x) \neq 0$ for all values of x is called Rational Fraction.

It has two types

Proper Rational Fraction	Improper Rational Fraction
A fraction is proper if degree of the polynomial in numerator is less than the degree of the polynomial in the denominator.	A fraction is improper if degree of the polynomial in numerator is greater than or equal to the degree of the polynomial in the denominator.

Partial Fractions:

“Partial Fractions” are rational fractions obtained by decomposing a single rational fraction into sum of two or more proper rational fractions.

Partial Fraction Resolution:

The process of decomposing a single rational fraction into sum of two or more proper rational fractions is called the “Partial Fraction Resolution”.

Note: There are two types of equations

Conditional Equation	Identity
An equation in which two algebraic expressions are equal for some particular value / values of the variable is called “Conditional equation”.	An equation in which two algebraic expressions are equal for all values of the variable is called the “Identity”.

Note:

- (i) For simplicity conditional equations will be called “equations” here.
- (ii) For both “equations” and “identities” “=” symbol will be used.
- (iii) Any improper fraction can be converted into a mixed form. Which consist of sum of a polynomial and a proper “rational fraction”.
- (iv) When a rational fraction is resolved and represented into partial fractions, the result is an identity.

Theorem:

Two polynomials are equal for all values of the variable if and only if these have the same degree and equal co-efficients for all like terms.

How to resolve a rational fraction $\frac{P(x)}{Q(x)}$:

- (i) If $\frac{P(x)}{Q(x)}$ is not a “proper fraction, then first write $\frac{P(x)}{Q(x)}$ into mixed form.
- (ii) Identify the type of given rational fraction.
- (iii) Write an identity according to the type, it will consist of some unknown constant.
- (iv) Simplify and equate the co-efficients of like terms on both sides of identity.
- (v) Solved the resulting equations to get value of unknown constants.

Case-I:

When $Q(x)$ has only non-repeated linear factors:

The polynomial $Q(x)$ may be written as:

$$Q(x) = (x - a_1)(x - a_2) \dots (x - a_n) \text{ where } a_1 \neq a_2 \neq \dots \neq a_n$$

$$\therefore \frac{P(x)}{Q(x)} = \frac{A_1}{x - a_1} + \frac{A_2}{x - a_2} + \dots + \frac{A_n}{x - a_n} \text{ is an identity.}$$

Where, the coefficients $A_1, A_2, \dots, A_n$ are unknown constants to be found.

Example:

Resolve $\frac{7x+25}{(x+3)(x+4)}$ into partial fractions.

Solution:

Suppose

$$\frac{7x+25}{(x+3)(x+4)} = \frac{A}{x+3} + \frac{B}{x+4} \quad (i)$$

Multiplying both sides $(x+3)(x+4)$

$$\Rightarrow 7x+25 = A(x+4) + B(x+3) \quad (ii)$$

As two sides of the identity are equal for all values of x ,

Let us put $x+3=0 \Rightarrow x=-3$ in equation (ii)

We get $-21+25 = A(-3+4)$

$$\Rightarrow A = 4$$

Putting $x+4=0 \Rightarrow x=-4$ in equation (ii)

We get $-28+25 = B(-4+3)$

$$\Rightarrow B = 3$$

Hence the partial fractions are:

$$\boxed{\frac{7x+25}{(x+3)(x+4)} = \frac{4}{x+3} + \frac{3}{x+4}}.$$

EXERCISE 5.1

Resolve the following into partial fractions.

Q.1 $\frac{1}{x^2 - 1}$

Solution: $\frac{1}{x^2 - 1} = \frac{1}{(x+1)(x-1)}$

Let, $\frac{1}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1}$ (i)

Q Multiplying both sides by $(x+1)(x-1)$, we get

$$\frac{1}{(x+1)(x-1)} \times (x+1)(x-1) = \left(\frac{A}{\cancel{x+1}} \right) (\cancel{x+1})(x-1) + \left(\frac{B}{\cancel{x-1}} \right) (x+1)(\cancel{x-1})$$

$$1 = A(x-1) + B(x+1) \quad \text{(ii)}$$

As two sides of identity are equal for all values of x , let us put

$x-1=0 \Rightarrow x=1$ in equation (ii), we get

$$1 = A(0) + B(2)$$

$$\Rightarrow \boxed{B = \frac{1}{2}}$$

Q Putting $x+1=0 \Rightarrow x=-1$ in equation (ii), we get

$$1 = A(-1-1) + B(-1+1)$$

$$1 = A(-2) + 0$$

$$\Rightarrow \boxed{A = -\frac{1}{2}}$$

Putting the values of A and B in equation (i)

So we have,

$$\frac{1}{(x+1)(x-1)} = \frac{-\frac{1}{2}}{x+1} + \frac{\frac{1}{2}}{x-1}$$

Hence the partial fractions are:

$$\boxed{\frac{1}{(x+1)(x-1)} = \frac{1}{2(x-1)} - \frac{1}{2(x+1)}}$$

Q.2 $\frac{x^2 + 1}{(x+1)(x-1)}$

Solution:

Since the degree of polynomial of numerator and denominator is equal so, it is improper Rational Fraction. First transform it into mixed form by division.

$$\frac{x^2 + 1}{x^2 - 1}$$

$$\begin{array}{r} x^2 - 1 \overline{) \cancel{x^2} + 1} \\ \underline{- \cancel{x^2} + 1} \\ 2 \end{array}$$

Dividing $x^2 + 1$ by $x^2 - 1$, we have quotient = 1 and remainder = 2, therefore

$$\frac{x^2 + 1}{x^2 - 1} = 1 + \frac{2}{x^2 - 1} = 1 + \frac{2}{(x+1)(x-1)} \quad (i)$$

$$\text{Let, } \frac{2}{(x+1)(x-1)} = \frac{A}{x+1} + \frac{B}{x-1} \quad (ii)$$

Multiplying by $(x+1)(x-1)$ on both sides of equation (ii), we get

$$2 = A(x-1) + B(x+1) \quad (iii)$$

Which is identity equation in x .

Putting $x-1=0 \Rightarrow x=-1$ in equation (iii)

$$2 = A(0) + B(1+1)$$

$$2 = 2B$$

$$\Rightarrow \boxed{B=1}$$

Putting $x+1=0 \Rightarrow x=-1$ in equation (iii)

$$2 = A(-1-1) + B(0)$$

$$2 = -2A$$

$$\Rightarrow \boxed{A=-1}$$

Putting the values of A and B in equation (ii), we get

$$\frac{2}{(x+1)(x-1)} = \frac{-1}{x+1} + \frac{1}{x-1}$$

By using equation (i)

Hence the partial fraction are:

$$\boxed{\frac{x^2 + 1}{x^2 - 1} = 1 + \frac{1}{x-1} - \frac{1}{x+1}}$$

Q.3
$$\frac{2x+1}{(x-1)(x+2)(x+3)}$$

Solution: Let,

$$\frac{2x+1}{(x-1)(x+2)(x+3)} = \frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{x+3} \quad (i)$$

Multiplying by $(x-1)(x+2)(x+3)$ on both sides of equation (i), we get

$$(2x+1) = A(x+2)(x+3) + B(x-1)(x+3) + C(x-1)(x+2) \quad (ii)$$

which is an identity in x

Putting $x-1=0 \Rightarrow x=1$ in equation (ii)

$$2(1)+1 = A(1+2)(1+3) + B(0)(1+3) + C(0)(1+2)$$

$$3 = A(3)(4) + 0 + 0$$

$$\Rightarrow 12A = 3$$

$$\Rightarrow A = \frac{3}{12}$$

$$\boxed{A = \frac{1}{4}}$$

Putting $x+2=0 \Rightarrow x=-2$ in equation (ii)

$$2(-2)+1 = A(0)(-2+3) + B(-2-1)(-2+3) + C(-2-1)(0)$$

$$-4+1 = 0 + B(-3)(1) + 0$$

$$-3 = -3B$$

$$\Rightarrow \boxed{B = 1}$$

Putting $x+3=0 \Rightarrow x=-3$ in equation (ii)

$$2(-3)+1 = A(-3+2)(0) + B(-3-1)(0) + C(-3-1)(-3+2)$$

$$-6+1 = 0 + 0 + C(-4)(-1)$$

$$-5 = 4C$$

$$\Rightarrow \boxed{C = -\frac{5}{4}}$$

Putting values of A , B and C in equation (i)

$$\frac{2x+1}{(x-1)(x+2)(x+3)} = \frac{\frac{1}{4}}{x-1} + \frac{1}{x+2} + \frac{-\frac{5}{4}}{x+3}$$

Hence the partial fractions are:

$$\boxed{\frac{2x+1}{(x-1)(x+2)(x+3)} = \frac{1}{4(x-1)} + \frac{1}{x+2} - \frac{5}{4(x+3)}}$$

Q.4
$$\frac{3x^2 - 4x - 5}{(x-2)(x^2 + 7x + 10)}$$

Solution: The factor $x^2 + 7x + 10$ in the denominator can be factorized, as

$$\text{Consider } x^2 + 7x + 10 = x^2 + 5x + 2x + 10$$

$$x(x+5) + 2(x+5) = (x+5)(x+2)$$

So,

$$\frac{3x^2 - 4x - 5}{(x-2)(x^2 + 7x + 10)} = \frac{3x^2 - 4x - 5}{(x-2)(x+2)(x+5)}$$

Let

$$\frac{3x^2 - 4x - 5}{(x-2)(x+2)(x+5)} = \frac{A}{x-2} + \frac{B}{x+2} + \frac{C}{x+5} \quad (i)$$

Multiplying $(x-2)(x+2)(x+5)$ on both sides of equation (i), we get

$$3x^2 - 4x - 5 = A(x+2)(x+5) + B(x-2)(x+5) + C(x-2)(x+2) \quad (ii)$$

Which is identity in x

Putting $x-2=0 \Rightarrow x=2$ in equation (ii)

$$3(2)^2 - 4(2) - 5 = A(2+2)(2+5) + B(0)(2+5) + C(0)(2+2)$$

$$3(4) - 8 - 5 = A(4)(7) + 0 + 0$$

$$-1 = 28A$$

$$\Rightarrow \boxed{A = \frac{-1}{28}}$$

Putting $x+2=0 \Rightarrow x=-2$ in equation (ii)

$$3(-2)^2 - 4(-2) - 5 = A(0)(-2+5) + B(-2-2)(-2+5) + C(-2-2)(0)$$

$$12 + 8 - 5 = 0 + B(-4)(3) + 0$$

$$15 = -12B$$

$$\Rightarrow B = \frac{-15}{12}$$

$$\boxed{B = -\frac{5}{4}}$$

Putting $x+5=0 \Rightarrow x=-5$ in equation (ii)

$$3(-5)^2 - 4(-5) - 5 = A(-5+2)(0) + B(-5-2)(0) + C(-5-2)(-5+2)$$

$$75 + 20 - 5 = 0 + 0 + C(-7)(-3)$$

$$90 = 21C$$

$$\Rightarrow C = \frac{90}{21}$$

$$\boxed{C = \frac{30}{7}}$$

Putting the values of A , B and C in equation (i)

$$\frac{3x^2 - 4x - 5}{(x-2)(x+2)(x+5)} = \frac{-\frac{1}{28}}{x-2} + \frac{-\frac{5}{4}}{x+2} + \frac{\frac{30}{7}}{x+5}$$

Hence the partial fraction are:

$$\boxed{\frac{3x^2 - 4x - 5}{(x-2)(x+2)(x+5)} = \frac{30}{7(x+5)} - \frac{1}{28(x-2)} - \frac{5}{4(x+2)}}$$

Q.5
$$\frac{1}{(x-1)(2x-1)(3x-1)}$$

Solution:

$$\text{Let } \frac{1}{(x-1)(2x-1)(3x-1)} = \frac{A}{x-1} + \frac{B}{2x-1} + \frac{C}{3x-1} \quad (i)$$

Multiplying by $(x-1)(2x-1)(3x-1)$ on both sides of equation (i), we get

$$1 = A(2x-1)(3x-1) + B(x-1)(3x-1) + C(x-1)(2x-1) \quad (ii)$$

Which is identity in x .

Putting $x-1=0 \Rightarrow x=1$ in equation (ii)

$$1 = A(2(1)-1)(3(1)-1) + B(0)(3(1)-1) + C(0)(2(1)-1)$$

$$1 = A(1)(2) + 0 + 0$$

$$\boxed{\Rightarrow A = \frac{1}{2}}$$

Putting $2x-1=0 \Rightarrow 2x=1 \Rightarrow x=\frac{1}{2}$ in equation (ii)

$$1 = A(0) \left[3\left(\frac{1}{2}\right) - 1 \right] + B\left(\frac{1}{2} - 1\right) \left[3\left(\frac{1}{2}\right) - 1 \right] + C\left(\frac{1}{2} - 1\right)(0)$$

$$1 = 0 + B\left(\frac{-1}{2}\right)\left(\frac{3-2}{2}\right) + 0$$

$$1 = B\left(\frac{-1}{2}\right)\left(\frac{1}{2}\right)$$

$$1 = B\left(\frac{-1}{4}\right)$$

$$\Rightarrow \boxed{B = -4}$$

Putting $3x - 1 = 0 \Rightarrow 3x = 1 \Rightarrow x = \frac{1}{3}$ in equation (ii), we get

$$1 = A\left[2\left(\frac{1}{3}\right) - 1\right](0) + B\left(\frac{1}{3} - 1\right)(0) + C\left(\frac{1}{3} - 1\right)\left[2\left(\frac{1}{3}\right) - 1\right]$$

$$1 = 0 + 0 + C\left(\frac{1-3}{3}\right)\left(\frac{2-3}{3}\right)$$

$$1 = C\left(\frac{-2}{3}\right)\left(\frac{-1}{3}\right)$$

$$1 = C\left(\frac{2}{9}\right)$$

$$\Rightarrow \boxed{C = \frac{9}{2}}$$

Putting the values of A, B and C in equation (i), we get

$$\frac{1}{(x-1)(2x-1)(3x-1)} = \frac{\frac{1}{2}}{x-1} + \frac{-4}{2x-1} + \frac{\frac{9}{2}}{3x-1}$$

Hence the partial fractions are:

$$\boxed{\frac{1}{(x-1)(2x-1)(3x-1)} = \frac{1}{2(x-1)} - \frac{4}{2x-1} + \frac{9}{2(3x-1)}}$$

Q.6
$$\frac{x}{(x-a)(x-b)(x-c)}$$

Solution: Let,

$$\frac{x}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c} \quad (i)$$

Multiplying $(x-a)(x-b)(x-c)$ on both sides of equation (i), we get

Which is identity in x

$$x = A(x-b)(x-c) + B(x-a)(x-c) + C(x-a)(x-b) \quad (ii)$$

Putting $x - a = 0 \Rightarrow x = a$ in equation (ii)

$$a = A(a-b)(a-c) + B(0)(a-c) + C(0)(a-b)$$

$$a = A(a-b)(a-c) + 0 + 0$$

$$\Rightarrow A = \frac{a}{(a-b)(a-c)}$$

Putting $x-b=0 \Rightarrow x=b$ in equation (ii), we get

$$b = A(0)(b-c) + B(b-a)(b-c) + C(b-a)(0)$$

$$b = 0 + B(b-a)(b-c) + 0$$

$$\Rightarrow B = \frac{b}{(b-a)(b-c)}$$

Putting $x-c=0 \Rightarrow x=c$ in equation (ii), we get

$$c = A(c-b)(0) + B(c-a)(0) + C(c-a)(c-b)$$

$$c = 0 + 0 + C(c-a)(c-b)$$

$$c = C(c-a)(c-b)$$

$$\Rightarrow C = \frac{c}{(c-a)(c-b)}$$

Putting the values of A , B and C in equation (i), we get

$$\frac{x}{(x-a)(x-b)(x-c)} = \frac{\frac{a}{(a-b)(a-c)}}{x-a} + \frac{\frac{b}{(b-a)(b-c)}}{x-b} + \frac{\frac{c}{(c-a)(c-b)}}{x-c}$$

Hence the partial fractions are:

$$\frac{x}{(x-a)(x-b)(x-c)} = \frac{a}{(x-a)(a-b)(a-c)} + \frac{b}{(x-b)(b-a)(b-c)} + \frac{c}{(x-c)(c-a)(c-b)}$$

Q.7
$$\frac{6x^3 + 5x^2 - 7}{2x^2 - x - 1}$$

Solution:

It is improper rational fraction so, transform it into mixed form.

$$\begin{array}{r} 3x+4 \\ 2x^2-x-1 \overline{) 6x^3+5x^2-7} \\ \underline{6x^3-3x^2-3x} \\ 8x^2+3x-7 \\ \underline{8x^2-4x-4} \\ 7x-3 \end{array}$$

Dividing $6x^3 + 5x^2 - 7$ by $2x^2 - x - 1$, we have

Quotient = $3x + 4$ and Remainder $7x - 3$.

$$\therefore \frac{6x^3 + 5x^2 - 7}{2x^2 - x - 1} = 3x + 4 + \frac{7x - 3}{2x^2 - x - 1} \quad (i)$$

Consider, $2x^2 - x - 1$

$$\Rightarrow 2x^2 - 2x + x - 1$$

$$\Rightarrow 2x(x-1) + 1(x-1) = (x-1)(2x+1)$$

So (i) becomes

$$\frac{6x^3 + 5x^2 - 7}{2x^2 - x - 1} = 3x + 4 + \frac{7x - 3}{(x-1)(2x+1)} \quad (\text{ii})$$

Let,

$$\frac{7x-3}{(x-1)(2x+1)} = \frac{A}{x-1} + \frac{B}{2x+1} \quad (\text{iii})$$

Multiplying $(x-1)(2x+1)$ on both sides of equation (iii), we get

$$7x-3 = A(2x+1) + B(x-1) \quad (\text{iv})$$

Which is identity in x .

Putting $x-1=0 \Rightarrow x=1$ in equation (iv)

$$7(1)-3 = A(2(1)+1) + B(0)$$

$$4 = 3A$$

$$\Rightarrow \boxed{A = \frac{4}{3}}$$

Putting $2x+1=0 \Rightarrow x=-\frac{1}{2}$ in equation (iv), we get

$$7\left(-\frac{1}{2}\right) - 3 = A(0) + B\left(-\frac{1}{2} - 1\right)$$

$$\frac{-7-6}{2} = B\left(-\frac{3}{2}\right)$$

$$\frac{-13}{2} = B\left(-\frac{3}{2}\right)$$

$$\Rightarrow B = \left(-\frac{13}{2}\right)\left(-\frac{2}{3}\right)$$

$$\boxed{B = \frac{13}{3}}$$

Putting the values of A and B in equation (iii), we get

$$\frac{7x-3}{(x-1)(2x+1)} = \frac{\frac{4}{3}}{x-1} + \frac{\frac{13}{3}}{2x+1}$$

By using equation (ii), we have

$$\boxed{\frac{6x^3 + 5x^2 - 7}{2x^2 - x - 1} = 3x + 4 + \frac{4}{3(x-1)} + \frac{13}{3(2x+1)}}$$

Q.8 $\frac{2x^3 + x^2 - 5x + 3}{2x^3 + x^2 - 3x}$

Solution:

It is improper rational fraction first transform it into mixed form.

$$2x^3 + x^2 - 3x \overline{) \begin{array}{r} 1 \\ 2x^3 + x^2 - 5x + 3 \\ \underline{2x^3 + x^2 - 3x} \\ -2x + 3 \end{array}}$$

Dividing $2x^3 + x^2 - 5x + 3$ by $2x^3 + x^2 - 3x$, we get

Quotient = 1 and Remainder = $-2x + 3$

$$\begin{aligned} \therefore \frac{2x^3 + x^2 - 5x + 3}{2x^3 + x^2 - 3x} &= 1 + \frac{-2x + 3}{2x^3 + x^2 - 3x} \\ &= 1 - \frac{2x - 3}{x(2x^2 + x - 3)} \end{aligned} \quad (i)$$

The factor in the denominator $2x^2 + x - 3$ can be factorized as,

$$\begin{aligned} 2x^2 + 3x - 2x - 3 \\ \Rightarrow x(2x + 3) - 1(2x + 3) \\ \Rightarrow (2x + 3)(x - 1) \end{aligned}$$

So (i) becomes,

$$\frac{2x^3 + x^2 - 5x + 3}{2x^3 + x^2 - 3x} = 1 - \frac{2x - 3}{x(x - 1)(2x + 3)} \quad (ii)$$

$$\text{Let, } \frac{2x - 3}{x(x - 1)(2x + 3)} = \frac{A}{x} + \frac{B}{x - 1} + \frac{C}{2x + 3} \quad (iii)$$

Multiplying by $x(x - 1)(2x + 3)$ on both sides of equation (iii), we get

$$2x - 3 = A(x - 1)(2x + 3) + B(2x + 3)(x) + C(x)(x - 1) \quad (iv)$$

Which is identity in x

Putting $x = 0$ in equation (iv)

$$2(0) - 3 = A(0 - 1)(2(0) + 3) + B(2(0) + 3)(0) + C(0)(0 - 1)$$

$$-3 = A(-1)(3) + 0 + 0$$

$$\Rightarrow -3A = -3$$

$$\boxed{A = 1}$$

Putting $x - 1 = 0 \Rightarrow x = 1$ in equation (iv)

$$2(1) - 3 = A(0)(2(1) + 3) + B(1)(2(1) + 3) + C(1)(0)$$

$$-1 = 0 + 5B + 0$$

$$\Rightarrow 5B = -1$$

$$\boxed{B = -\frac{1}{5}}$$

Putting $2x + 3 = 0 \Rightarrow 2x = -3 \Rightarrow x = -\frac{3}{2}$ in equation (iv), we get

$$2\left(\frac{-3}{2}\right) - 3 = A\left(\frac{-3}{2} - 1\right)(0) + B(0)\left(-\frac{3}{2}\right) + C\left(\frac{-3}{2}\right)\left(\frac{-3}{2} - 1\right)$$

$$-6 = 0 + 0 + C\left(\frac{-3}{2}\right)\left(\frac{-5}{2}\right)$$

$$-6 = \frac{15}{2}C$$

$$\Rightarrow C = (-6)\left(\frac{2}{15}\right)$$

$$\boxed{C = -\frac{4}{5}}$$

Putting the values of A , B and C in equation (iii), we get

$$\frac{2x-3}{x(x-1)(2x+3)} = \frac{1}{x} + \frac{-\frac{1}{5}}{x-1} + \frac{-\frac{4}{5}}{2x+3} \quad \text{by using equation (i), we get}$$

$$\frac{2x^3 + x^2 - 5x + 3}{2x^3 + x^2 - 3x} = 1 - \left[\frac{1}{x} - \frac{1}{5(x-1)} - \frac{4}{5(2x+3)} \right]$$

Hence the partial fractions are:

$$\boxed{\frac{2x^3 + x^2 - 5x + 3}{2x^3 + x^2 - 3x} = 1 - \frac{1}{x} + \frac{1}{5(x-1)} + \frac{4}{5(2x+3)}}$$

Q.9
$$\frac{(x-1)(x-3)(x-5)}{(x-2)(x-4)(x-6)}$$

Solution: It is improper rational fraction first transform it into mixed form.

$$\begin{aligned} \frac{(x-1)(x-3)(x-5)}{(x-2)(x-4)(x-6)} &= \frac{(x-1)(x^2-8x+15)}{(x-2)(x^2-10x+24)} \\ &= \frac{x^3-8x^2+15x-x^2+8x-15}{x^3-10x^2+24x-2x^2+20x-48} \\ &= \frac{x^3-9x^2+23x-15}{x^3-12x^2+44x-48} \end{aligned}$$

$$\begin{array}{r} x^3 - 12x^2 + 44x - 48 \overline{) \frac{1}{x^3 - 9x^2 + 23x - 15}} \\ \underline{x^3 - 12x^2 + 44x - 48} \\ 3x^2 - 21x + 33 \end{array}$$

So,

Dividing $x^3 - 9x^2 + 23x - 15$ by $x^3 - 12x^2 + 44x - 48$, we have

Quotient = 1 and Remainder = $3x^2 - 21x + 33$

$$\frac{x^3 - 9x^2 + 23x - 15}{x^3 - 12x^2 + 44x - 48} = 1 + \frac{3x^2 - 21x + 33}{x^3 - 12x^2 + 44x - 48} = 1 + \frac{3x^2 - 21x + 33}{(x-2)(x-4)(x-6)} \quad (i)$$

Let,

$$\frac{3x^2 - 21x + 33}{(x-2)(x-4)(x-6)} = \frac{A}{x-2} + \frac{B}{x-4} + \frac{C}{x-6} \quad (ii)$$

Multiplying by $(x-2)(x-4)(x-6)$ to equation (ii)

$$3x^2 - 21x + 33 = A(x-4)(x-6) + B(x-2)(x-6) + C(x-2)(x-4) \quad (iii)$$

Putting $x-2=0 \Rightarrow x=2$ in equation (iii), we have

$$3(2)^2 - 21(2) + 33 = A(2-4)(2-6) + B(0)(2-6) + C(0)(2-4)$$

$$12 - 42 + 33 = A(-2)(-4) + 0 + 0$$

$$3 = 8A$$

$$\Rightarrow A = \frac{3}{8}$$

Putting $x-4=0 \Rightarrow x=4$ in equation (iii)

$$3(4)^2 - 21(4) + 33 = A(0)(4-6) + B(4-2)(4-6) + C(4-2)(0)$$

$$48 - 84 + 33 = 0 + B(2)(-2) + 0$$

$$-3 = -4B$$

$$\Rightarrow B = \frac{-3}{-4}$$

$$B = \frac{3}{4}$$

Putting $x-6=0 \Rightarrow x=6$ in equation (iii)

$$3(6)^2 - 21(6) + 33 = A(6-4)(0) + B(6-2)(0) + C(6-2)(6-4)$$

$$3(36) - 126 + 33 = 0 + 0 + C(4)(2)$$

$$108 - 126 + 33 = 8C$$

$$15 = 8C$$

$$\Rightarrow C = \frac{15}{8}$$

Putting the values of A, B and C in equation (ii), we get

$$\frac{3x^2 - 21x + 33}{(x-2)(x-4)(x-6)} = \frac{\frac{3}{8}}{x-2} + \frac{\frac{3}{4}}{x-4} + \frac{\frac{15}{8}}{x-6}$$

By using (i), partial fractions are:

$$\frac{(x-1)(x-3)(x-5)}{(x-2)(x-4)(x-6)} = 1 + \frac{3}{8(x-2)} + \frac{3}{4(x-4)} + \frac{15}{8(x-6)}$$

Q.10
$$\frac{1}{(1-ax)(1-bx)(1-cx)}$$

Solution: Let

$$\frac{1}{(1-ax)(1-bx)(1-cx)} = \frac{A}{1-ax} + \frac{B}{1-bx} + \frac{C}{1-cx} \quad (i)$$

Multiplying by $(1-ax)(1-bx)(1-cx)$ on both sides of equation (i), we get

$$1 = A(1-bx)(1-cx) + B(1-ax)(1-cx) + C(1-ax)(1-bx) \quad (ii)$$

Putting $1-ax=0 \Rightarrow ax=1 \Rightarrow x=\frac{1}{a}$ in equation (ii)

$$1 = A\left[1-b\left(\frac{1}{a}\right)\right]\left[1-c\left(\frac{1}{a}\right)\right] + B(0)\left[1-c\left(\frac{1}{a}\right)\right] + C(0)\left[1-b\left(\frac{1}{a}\right)\right]$$

$$1 = A\left(1-\frac{b}{a}\right)\left(1-\frac{c}{a}\right) + 0 + 0$$

$$1 = A\left(\frac{a-b}{a}\right)\left(\frac{a-c}{a}\right)$$

$$\Rightarrow \boxed{A = \frac{a^2}{(a-b)(a-c)}}$$

Putting $1-bx=0 \Rightarrow bx=1 \Rightarrow x=\frac{1}{b}$ in equation (ii)

$$1 = A\left[1-c\left(\frac{1}{b}\right)\right](0) + B\left[1-a\left(\frac{1}{b}\right)\right]\left[1-c\left(\frac{1}{b}\right)\right] + c\left[1-a\left(\frac{1}{b}\right)\right](0)$$

$$1 = 0 + B\left(1-\frac{a}{b}\right)\left(1-\frac{c}{b}\right) + 0$$

$$1 = B\left(\frac{b-a}{b}\right)\left(\frac{b-c}{b}\right)$$

$$\Rightarrow \boxed{B = \frac{b^2}{(b-a)(b-c)}}$$

Putting $1-cx=0 \Rightarrow cx=1 \Rightarrow x=\frac{1}{c}$ in equation (ii)

$$1 = A\left[1-b\left(\frac{1}{c}\right)\right](0) + B\left[1-a\left(\frac{1}{c}\right)\right](0) + C\left[1-a\left(\frac{1}{c}\right)\right]\left[1-b\left(\frac{1}{c}\right)\right]$$

$$1 = 0 + 0 + C\left(1-\frac{a}{c}\right)\left(1-\frac{b}{c}\right)$$

$$1 = C\left(\frac{c-a}{c}\right)\left(\frac{c-b}{c}\right)$$

$$\Rightarrow C = \frac{c^2}{(c-a)(c-b)}$$

Putting the values of A, B and C in equation (i)

$$\frac{1}{(1-ax)(1-bx)(1-cx)} = \frac{a^2 / (a-b)(a-c)}{1-ax} + \frac{b^2 / (b-a)(b-c)}{1-bx} + \frac{c^2 / (c-a)(c-b)}{1-cx}$$

Hence the partial fractions are:

$$\frac{1}{(1-ax)(1-bx)(1-cx)} = \frac{a^2}{(a-b)(a-c)(1-ax)} + \frac{b^2}{(b-a)(b-c)(1-bx)} + \frac{c^2}{(c-a)(c-b)(1-cx)}$$

Q.11
$$\frac{x^2 + a^2}{(x^2 + b^2)(x^2 + c^2)(x^2 + d^2)}$$

Solution: Let,

$$x^2 = y, \text{ then}$$

$$\frac{y + a^2}{(y + b^2)(y + c^2)(y + d^2)}$$

Let,

$$\frac{y + a^2}{(y + b^2)(y + c^2)(y + d^2)} = \frac{A}{y + b^2} + \frac{B}{y + c^2} + \frac{C}{y + d^2} \quad (i)$$

Multiplying by $(y + b^2)(y + c^2)(y + d^2)$ on both sides of equation (i) we get

$$y + a^2 = A(y + c^2)(y + d^2) + B(y + b^2)(y + d^2) + C(y + b^2)(y + c^2) \quad (ii)$$

Putting $y + b^2 = 0 \Rightarrow y = -b^2$ in equation (ii)

$$-b^2 + a^2 = A(-b^2 + c^2)(-b^2 + d^2) + B(0)(-b^2 + d^2) + C(0)(-b^2 + c^2)$$

$$a^2 - b^2 = A(c^2 - b^2)(d^2 - b^2) + 0 + 0$$

$$\Rightarrow A = \frac{a^2 - b^2}{(c^2 - b^2)(d^2 - b^2)}$$

Putting $y + c^2 = 0 \Rightarrow y = -c^2$ in equation (ii)

$$-c^2 + a^2 = A(0)(-c^2 + d^2) + B(-c^2 + b^2)(-c^2 + d^2) + C(-c^2 + b^2)(0)$$

$$a^2 - c^2 = 0 + B(b^2 - c^2)(d^2 - c^2) + 0$$

$$\Rightarrow B = \frac{a^2 - c^2}{(b^2 - c^2)(d^2 - c^2)}$$

Putting $y + d^2 = 0 \Rightarrow y = -d^2$ in equation (ii), we get

$$-d^2 + a^2 = A(-d^2 + c^2)(0) + B(-d^2 + b^2)(0) + C(-d^2 + b^2)(-d^2 + c^2)$$

$$a^2 - d^2 = C(b^2 - d^2)(c^2 - d^2)$$

$$\Rightarrow C = \frac{a^2 - d^2}{(b^2 - d^2)(c^2 - d^2)}$$

Putting the values of A, B and C in equation (i), we get

$$\begin{aligned} \frac{y + a^2}{(y + b^2)(y + c^2)(y + d^2)} &= \frac{(a^2 - b^2) / (c^2 - b^2)(d^2 - b^2)}{y + b^2} + \frac{(a^2 - c^2) / (b^2 - c^2)(d^2 - c^2)}{y + c^2} \\ &+ \frac{(a^2 - d^2) / (b^2 - d^2)(c^2 - d^2)}{y + d^2} \end{aligned}$$

Since $y = x^2$

So,

Partial fractions are

$$\begin{aligned} \frac{x^2 + a^2}{(x^2 + b^2)(x^2 + c^2)(x^2 + d^2)} &= \frac{a^2 - b^2}{(x^2 + b^2)(c^2 - b^2)(d^2 - b^2)} + \frac{a^2 - c^2}{(x^2 + c^2)(b^2 - c^2)(d^2 - c^2)} \\ &+ \frac{a^2 - d^2}{(x^2 + d^2)(b^2 - d^2)(c^2 - d^2)} \end{aligned}$$

Case-II:

When $Q(x)$ has repeated linear factors:

If the polynomial has a factor $(x-a)^n, n \geq 2$ and n is a positive integer, then $\frac{P(x)}{Q(x)}$ may

be written as the following identity:

$$\therefore \frac{P(x)}{Q(x)} = \frac{A_1}{(x-a_1)} + \frac{A_2}{(x-a_2)^2} + \dots + \frac{A_n}{(x-a_n)^n}$$

Where, the coefficients $A_1, A_2, \dots, A_n$ are unknown constants to be found.

Example:

Resolve $\frac{x^2 + x - 1}{(x+2)^3}$ into partial fractions.

Solution:

Suppose

$$\frac{x^2 + x - 1}{(x+2)^3} = \frac{A}{x+2} + \frac{B}{(x+2)^2} + \frac{C}{(x+2)^3} \quad (i)$$

Multiplying both sides $(x+2)^3$

$$\Rightarrow x^2 + x - 1 = A(x+2)^2 + B(x+2) + C \quad (ii)$$

$$\Rightarrow x^2 + x - 1 = A(x^2 + 4x + 4) + B(x+2) + C \quad (iii)$$

Let us put $x+2=0 \Rightarrow x=-2$ in equation (ii)

We get $(-2)^2 + (-2) - 1 = A(0) + B(0) + C$

$$\Rightarrow 1 = C$$

Comparing the coefficients of x^2 in equation (iii)

We get $A=1$ and

Comparing the coefficients of x in equation (iii)

We get $1 = 4A + B$

$$\Rightarrow 1 = 4 + B$$

$$\Rightarrow B = -3$$

Hence the partial fractions are:

$$\boxed{\frac{x^2 + x - 1}{(x+2)^3} = \frac{1}{x+2} - \frac{3}{(x+2)^2} + \frac{1}{(x+2)^3}}.$$

EXERCISE 5.2

Q:1 Resolve $\frac{2x^2 - 3x + 4}{(x-1)^3}$ into Partial Fraction .

Solution:

$$\text{Let } \frac{2x^2 - 3x + 4}{(x-1)^3} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3} \quad (i)$$

Multiplying by $(x-1)^3$ on both sides of equation (i), we get

$$2x^2 - 3x + 4 = A(x-1)^2 + B(x-1) + C \quad (ii)$$

Putting $x-1=0 \Rightarrow x=1$ in equation (ii), we get

$$2(1)^2 - 3(1) + 4 = A(0)^2 + B(0) + C$$

$$2 - 3 + 4 = 0 + 0 + C$$

$$\boxed{C=3}$$

Expanding equation (ii), we get

$$2x^2 - 3x + 4 = A(x^2 + 1 - 2x) + B(x-1) + C$$

$$2x^2 - 3x + 4 = Ax^2 + A - 2Ax + Bx - B + C$$

$$2x^2 - 3x + 4 = Ax^2 + (B - 2A)x + A - B + C \quad (iii)$$

Comparing the coefficients of x^2 from equation (iii), we get

$$\boxed{A=2}$$

Similarly comparing coefficients of x from equation (iii), we get

$$-3 = B - 2A$$

$$2A - 3 = B$$

$$\Rightarrow B = 2A - 3$$

$$B = 2(2) - 3 \quad \because A=2$$

$$\boxed{B=1}$$

Putting the values of A , B and C in equation (i), we get

$$\boxed{\frac{2x^2 - 3x + 4}{(x-1)^3} = \frac{2}{x-1} + \frac{1}{(x-1)^2} + \frac{3}{(x-1)^3}}$$

Q:2 Resolve, $\frac{5x^2 - 2x + 3}{(x+2)^3}$ into partial fractions.

Solution:

$$\text{Let } \frac{5x^2 - 2x + 3}{(x+2)^3} = \frac{A}{x+2} + \frac{B}{(x+2)^2} + \frac{C}{(x+2)^3} \quad (i)$$

Multiplying by $(x+2)^3$ on both sides of equation (i), we get

$$5x^2 - 2x + 3 = A(x+2)^2 + B(x+2) + C \quad (ii)$$

Putting $x+2=0 \Rightarrow x=-2$ in equation (ii), we get

$$5(-2)^2 - 2(-2) + 3 = A(0)^2 + B(0) + C$$

$$20 + 4 + 3 = 0 + 0 + C$$

$$\Rightarrow \boxed{C = 27}$$

Expanding equation (ii), we get

$$5x^2 - 2x + 3 = A(x^2 + 4 + 4x) + B(x+2) + C$$

$$5x^2 - 2x + 3 = Ax^2 + 4A + 4Ax + Bx + 2B + C$$

$$5x^2 - 2x + 3 = Ax^2 + (4A+B)x + 4A + 2B + C \quad (iii)$$

Comparing Coefficients of x^2 from equation (iii), we get

$$5 = A$$

$$\Rightarrow \boxed{A = 5}$$

Comparing Coefficients of x from equation (iii), we get

$$-2 = 4A + B$$

$$B = -2 - 4A$$

$$B = -2 - 4(5) \quad \because A=5$$

$$\boxed{B = -22}$$

Putting the values of A , B and C in equation (i), we get

$$\boxed{\frac{5x^2 - 2x + 3}{(x+2)^3} = \frac{5}{x+2} - \frac{22}{(x+2)^2} + \frac{27}{(x+2)^3}}$$

Q:3 Resolve, $\frac{4x}{(x+1)^2(x-1)}$ into Partial Fractions.

Solution:

Let
$$\frac{4x}{(x+1)^2(x-1)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-1} \quad (i)$$

Multiplying by $(x+1)^2(x-1)$ on both sides of equation (i), we get

$$4x = A(x+1)(x-1) + B(x-1) + C(x+1)^2 \quad (ii)$$

Putting $x+1=0 \Rightarrow x=-1$ in equation (ii), we get

$$4(-1) = A(0)(-1-1) + B(-1-1) + C(0)^2$$

$$-4 = 0 + B(-2) + 0$$

$$\Rightarrow -2B = -4$$

$$\boxed{B = 2}$$

Putting $x-1=0 \Rightarrow x=1$ in equation (ii), we get

$$4(1) = A(1+1)(0) + B(0) + C(1+1)^2$$

$$4 = 0 + 0 + 4C$$

$$\Rightarrow 4C = 4$$

$$C = \frac{4}{4}$$

$$\boxed{C = 1}$$

Expanding equation (ii), we get

$$4x = A(x^2 - 1) + B(x-1) + C(x^2 + 2x + 1)$$

$$4x = Ax^2 - A + Bx - B + Cx^2 + 2Cx + C$$

$$4x = Ax^2 + Cx^2 + 2Cx + Bx - A + C \quad (iii)$$

Comparing Coefficients of x^2 from equation (iii), we get

$$A + C = 0$$

$$A = -C$$

$$\boxed{A = -1} \quad \because C = 1$$

Putting the values of A, B and C in equation (i), we get

$$\frac{4x}{(x+1)^2(x-1)} = \frac{-1}{x+1} + \frac{2}{(x+1)^2} + \frac{1}{x-1}$$

$$\boxed{\frac{4x}{(x+1)^2(x-1)} = \frac{1}{x-1} - \frac{1}{x+1} + \frac{2}{(x+1)^2}}$$

Q:4 Resolve, $\frac{9}{(x+2)^2(x-1)}$ into Partial Fractions.

Solution:

$$\text{Let } \frac{9}{(x+2)^2(x-1)} = \frac{A}{(x+2)} + \frac{B}{(x+2)^2} + \frac{C}{(x-1)} \quad (i)$$

Multiplying by $(x+2)^2(x-1)$ on both sides of equation (i), we get

$$9 = A(x+2)(x-1) + B(x-1) + C(x+2)^2 \quad (ii)$$

Putting $x + 2 = 0 \Rightarrow x = -2$ in equation (ii), we get

$$9 = A(0)(-2-1) + B(-2-1) + C(0)^2$$

$$9 = 0 - 3B + 0$$

$$\Rightarrow -3B = 9 \Rightarrow B = -\frac{9}{3}$$

$$\boxed{B = -3}$$

Putting $x - 1 = 0 \Rightarrow x = 1$ in equation (ii), we get

$$9 = A(1+2)(0) + B(0) + C(1+2)^2$$

$$9 = 0 + 0 + 9C$$

$$\Rightarrow 9C = 9 \Rightarrow C = \frac{9}{9}$$

$$\boxed{C = 1}$$

Expanding equation (ii), we get

$$9 = A(x^2 + x - 2) + B(x - 1) + C(x^2 + 4x + 4)$$

$$9 = Ax^2 + Ax - 2A + Bx - B + Cx^2 + 4Cx + 4C$$

$$0x^2 + 0x + 9 = (A + C)x^2 + (A + B + 4C)x - 2A - B + 4C \quad (iii)$$

Comparing Coefficients of x^2 from equation (iii), we get

$$0 = A + C$$

$$\Rightarrow A + C = 0$$

$$A = -C$$

$$\boxed{A = -1} \quad \because C = 1$$

Putting the values of A , B and C in equation (i), we get

$$\frac{9}{(x+2)^2(x-1)} = \frac{-1}{x+2} + \frac{-3}{(x+2)^2} + \frac{1}{x-1}$$

$$\boxed{\frac{9}{(x+2)^2(x-1)} = \frac{1}{x-1} - \frac{1}{x+2} - \frac{3}{(x+2)^2}}$$

Q:5 Resolve, $\frac{1}{(x-3)^2(x+1)}$ into partial fractions.

Solution:

Let
$$\frac{1}{(x-3)^2(x+1)} = \frac{A}{x-3} + \frac{B}{(x-3)^2} + \frac{C}{x+1} \quad (i)$$

Multiplying by $(x-3)^2(x+1)$ on both side of equation (i), we get

$$1 = A(x-3)(x+1) + B(x+1) + C(x-3)^2 \quad (ii)$$

Putting $x-3=0 \Rightarrow x=3$ in equation (ii), we get

$$1 = A(0)(3+1) + B(3+1) + C(0)$$

$$1 = 0 + 4B + 0$$

$$\Rightarrow 4B = 1$$

$$\boxed{B = \frac{1}{4}}$$

Putting $x+1=0 \Rightarrow x=-1$ in equation (ii), we get

$$1 = A(-1-3)(0) + B(0) + C(-1-3)^2$$

$$1 = 0 + 0 + 16C$$

$$\Rightarrow 16C = 1$$

$$\boxed{C = \frac{1}{16}}$$

By expanding equation (ii), we get

$$1 = A(x^2 + x - 3x - 3) + B(x+1) + C(x^2 + 9 - 6x)$$

$$1 = A(x^2 - 2x - 3) + B(x+1) + C(x^2 - 6x + 9)$$

$$1 = Ax^2 - 2Ax - 3A + Bx + B + Cx^2 - 6Cx + 9C$$

$$1 = (A+C)x^2 + (B-2A-6C)x - 3A + B + 9C$$

Comparing Coefficient of x^2 , we get

$$0 = A + C$$

$$A = -C$$

$$\boxed{A = -\frac{1}{16}} \quad \therefore C = \frac{1}{16}$$

Putting the values of A, B and C in equation (i), we get

$$\frac{1}{(x-3)^2(x+1)} = -\frac{1}{16(x+1)} + \frac{1}{4(x-3)^2} + \frac{1}{16(x-3)}$$

$$\boxed{\frac{1}{(x-3)^2(x+1)} = \frac{1}{16(x+1)} - \frac{1}{16(x-3)} + \frac{1}{4(x-3)^2}}$$

Q:6 Resolve, $\frac{x^2}{(x-2)(x-1)^2}$ into partial fractions.

Solution:

Let
$$\frac{x^2}{(x-2)(x-1)^2} = \frac{A}{x-2} + \frac{B}{x-1} + \frac{C}{(x-1)^2} \quad (i)$$

Multiplying by $(x-2)(x-1)^2$ on both sides of equation (i), we get

$$x^2 = A(x-1)^2 + B(x-2)(x-1) + C(x-2) \quad (ii)$$

Putting $x-2=0 \Rightarrow x=2$ in equation (ii), we get

$$(2)^2 = A(2-1) + B(0)(2-1) + C(0)$$

$$4 = A(1) + 0 + 0$$

$$\Rightarrow \boxed{A=4}$$

Putting $x-1=0 \Rightarrow x=1$ in equation (ii), we get

$$(1)^2 = A(1-1)^2 + B(1-2)(0) + C(1-2)$$

$$1 = -C$$

$$\Rightarrow \boxed{C=-1}$$

By expanding equation (ii), we get

$$x^2 = A(x^2 + 1 - 2x) + B(x^2 - x - 2x + 2) + C(x-2)$$

$$x^2 = A(x^2 - 2x + 1) + B(x^2 - 3x + 2) + C(x-2)$$

$$x^2 = Ax^2 - 2Ax + A + Bx^2 - 3Bx + 2B + Cx - 2C$$

$$x^2 + 0x + 0 = (A+B)x^2 + (C-2A-3B)x + A+2B-2C$$

Comparing coefficient of x^2 , we get

$$1 = A + B$$

$$\Rightarrow B = 1 - A$$

$$B = 1 - 4 \quad \therefore A = 4$$

$$\boxed{B=-3}$$

Putting the values of A, B and C in equation (i), we get

$$\frac{x^2}{(x-2)(x-1)^2} = \frac{4}{x-2} + \frac{-3}{x-1} + \frac{-1}{(x-1)^2}$$

$$\boxed{\frac{x^2}{(x-2)(x-1)^2} = \frac{4}{x-2} - \frac{3}{x-1} - \frac{1}{(x-1)^2}}$$

Q:7 Resolve, $\frac{1}{(x-1)^2(x+1)}$ into partial Fractions.

Solution:

Let
$$\frac{1}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1} \quad (i)$$

Multiplying by $(x-1)^2(x+1)$ on both sides of equation (i), we get

$$1 = A(x-1)(x+1) + B(x+1) + C(x-1)^2 \quad (ii)$$

Putting $x-1=0 \Rightarrow x=1$ in equation (ii), we get

$$1 = A(0)(1+1) + B(1+1) + C(0)^2$$

$$1 = 0 + 2B + 0$$

$$\Rightarrow 2B = 1$$

$$\boxed{B = \frac{1}{2}}$$

Putting $x+1=0 \Rightarrow x=-1$ in equation (ii), we get

$$1 = A(-1-1)(0) + B(0) + C(-1-1)^2$$

$$1 = 0 + 0 + 4C$$

$$\Rightarrow 4C = 1$$

$$\boxed{C = \frac{1}{4}}$$

Expanding equation (ii), we get

$$1 = A(x^2-1) + B(x+1) + C(x^2+1-2x)$$

$$1 = Ax^2 - A + Bx + B + Cx^2 - 2Cx + C$$

$$0x^2 + 0x + 1 = (A+C)x^2 + (B-2C)x - A+B+C \quad (iii)$$

Comparing Coefficients of x^2 from equation (iii), we get

$$0 = A + C$$

$$\Rightarrow A + C = 0$$

$$A = -C$$

$$\boxed{A = -\frac{1}{4}} \quad \because C = \frac{1}{4}$$

Putting the values of A, B and C in equation (i), we get

$$\frac{1}{(x-1)^2(x+1)} = \frac{-\frac{1}{4}}{x-1} + \frac{\frac{1}{2}}{(x-1)^2} + \frac{\frac{1}{4}}{x+1}$$

$$\boxed{\frac{1}{(x-1)^2(x+1)} = \frac{1}{4(x+1)} - \frac{1}{4(x-1)} + \frac{1}{2(x-1)^2}}$$

Q:8 Resolve, $\frac{x^2}{(x-1)^3(x+1)}$ into Partial Fraction .

Solution:

Let
$$\frac{x^2}{(x-1)^3(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3} + \frac{D}{x+1} \quad (i)$$

Multiplying by $(x-1)^3(x+1)$ on both sides of equation (i), we get.

$$x^2 = A(x-1)^2(x+1) + B(x-1)(x+1) + C(x+1) + D(x-1)^3 \quad (ii)$$

Putting $x-1=0 \Rightarrow x=1$ in equation (ii), we get

$$(1)^2 = A(0)^2(1+1) + B(0)(1+1) + C(1+1) + D(0)^3$$

$$1 = 0 + 0 + 2C + 0$$

$$\Rightarrow 2C = 1$$

$$\boxed{C = \frac{1}{2}}$$

Putting $x+1=0 \Rightarrow x=-1$ in equation (ii), we get

$$(-1)^2 = A(-1-1)^2(0) + B(-1-1)(0) + C(0) + D(-1-1)^3$$

$$1 = 0 + 0 + 0 - 8D$$

$$\Rightarrow -8D = 1$$

$$\boxed{D = -\frac{1}{8}}$$

Expanding equation (ii), we get

$$x^2 = A(x^2 + 1 - 2x)(x+1) + B(x^2 - 1) + C(x+1) + D[x^3 - 1 - 3x(x-1)]$$

$$x^2 = A(x^3 + x^2 + x + 1 - 2x^2 - 2x) + B(x^2 - 1) + C(x+1) + D(x^3 - 1 - 3x^2 + 3x)$$

$$x^2 = A(x^3 - x^2 - x + 1) + B(x^2 - 1) + C(x+1) + D(x^3 - 3x^2 + 3x - 1)$$

$$0x^3 + x^2 + 0x + 0 = (A+D)x^3 + (B-A-3D)x^2 + (C-A+3D)x + (A-B+C-D) \quad (iii)$$

Comparing Coefficients of x^3 from equation (iii), we get

$$0 = A + D$$

$$\Rightarrow A + D = 0 \Rightarrow A = -D$$

$$A = -\left(-\frac{1}{8}\right) \quad \therefore D = -\frac{1}{8}$$

$$A = \frac{1}{8}$$

Comparing coefficients of x^2 from equation (iii), we get

$$1 = B - A - 3D$$

$$B = 1 + A + 3D$$

$$B = 1 + \frac{1}{8} - \frac{3}{8} \quad \therefore A = \frac{1}{8}, D = -\frac{1}{8}$$

$$B = \frac{8+1-3}{8}$$

$$B = \frac{6}{8}$$

$$\boxed{B = \frac{3}{4}}$$

Putting the values of A , B , C and D in equation (i), we get

$$\frac{x^2}{(x-1)^3(x+1)} = \frac{\frac{1}{8}}{x-1} + \frac{\frac{3}{4}}{(x-1)^2} + \frac{\frac{1}{2}}{(x-1)^3} + \frac{-\frac{1}{8}}{x+1}$$

$$\boxed{\frac{x^2}{(x-1)^3(x+1)} = \frac{1}{8(x-1)} + \frac{3}{4(x-1)^2} + \frac{1}{2(x-1)^3} - \frac{1}{8(x+1)}}$$

Q:9 Resolve, $\frac{x-1}{(x-2)(x+1)^3}$ into Partial Fractions.

Solution:

$$\text{Let } \frac{x-1}{(x-2)(x+1)^3} = \frac{A}{x-2} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3} \quad (i)$$

Multiplying by $(x-2)(x+1)^3$ on both sides of equation (i), we get

$$x-1 = A(x+1)^3 + B(x-2)(x+1)^2 + C(x-2)(x+1) + D(x-2) \quad (ii)$$

Putting $x-2 = 0 \Rightarrow x=2$ in equation (ii), we get

$$2-1 = A(2+1)^3 + B(0)(2+1)^2 + C(0)(2+1) + D(0)$$

$$1 = A(27) + 0 + 0 + 0$$

$$\Rightarrow 27A = 1$$

$$\boxed{A = \frac{1}{27}}$$

Putting $x+1=0 \Rightarrow x=-1$ in equation (ii), we get

$$-1-1 = A(0)^3 + B(-1-2)(0)^2 + C(-1-2)(0) + D(-1-2)$$

$$-2 = 0 + 0 + 0 + D(-3)$$

$$\Rightarrow -3D = -2$$

$$D = \frac{-2}{-3}$$

$$\boxed{D = \frac{2}{3}}$$

By Expanding equation (ii), we get

$$x-1 = A(x^3 + 1 + 3x^2 + 3x) + B(x-2)(x^2 + 2x + 1) + C(x^2 + x - 2x - 2) + D(x-2)$$

$$x-1 = A(x^3 + 3x^2 + 3x + 1) + B(x^3 + 2x^2 + x - 2x^2 - 4x - 2) + C(x^2 - x - 2) + D(x-2)$$

$$x-1 = A(x^3 + 3x^2 + 3x + 1) + B(x^3 - 3x - 2) + C(x^2 - x - 2) + D(x-2)$$

$$x-1 = Ax^3 + 3Ax^2 + 3Ax + A + Bx^3 - 3Bx - 2B + Cx^2 - Cx - 2C + Dx - 2D$$

Arranging terms of x^3, x^2, x and constant, we have

$$0x^3 + 0x^2 + x - 1 = (A+B)x^3 + (3A+C)x^2 + (3A-3B-C+D)x + A-2B-2C-2D \quad \text{(iii)}$$

Comparing the coefficient x^3 we have

$$0 = A + B$$

$$\Rightarrow B = -A$$

$$\boxed{B = -\frac{1}{27}}$$

$$\therefore A = \frac{1}{27}$$

Comparing Coefficients of x^2 from equation (iii), we get

$$0 = 3A + C$$

$$\Rightarrow C = -3A$$

$$C = -3\left(\frac{1}{27}\right)$$

$$\boxed{C = -\frac{1}{9}}$$

Putting the values of A, B, C and D in equation (i), we get

$$\frac{x-1}{(x-2)(x+1)^3} = \frac{\frac{1}{27}}{x-2} + \frac{-\frac{1}{27}}{x+1} + \frac{-\frac{1}{9}}{(x+1)^2} + \frac{\frac{2}{3}}{(x+1)^3}$$

$$\boxed{\frac{x-1}{(x-2)(x+1)^3} = \frac{1}{27(x-2)} - \frac{1}{27(x+1)} - \frac{1}{9(x+1)^2} + \frac{2}{3(x+1)^3}}$$

Q:10 Resolve, $\frac{4x^3}{(x^2-1)(x+1)^2}$ into Partial Fractions.

Solution:

$$\frac{4x^3}{(x^2-1)(x+1)^3} = \frac{4x^3}{(x+1)(x-1)(x+1)^2}$$

Let
$$\frac{4x^3}{(x-1)(x+1)^3} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3} \quad (i)$$

Multiplying by $(x-1)(x+1)^3$ on both sides of equation (i), we get

$$4x^3 = A(x+1)^3 + B(x-1)(x+1)^2 + C(x-1)(x+1) + D(x-1) \quad (ii)$$

Putting $x+1=0 \Rightarrow x=-1$ in equation (ii), we get

$$4(-1)^3 = A(0)^3 + B(-1-1)(0)^2 + C(-1-1)(0) + D(-1-1)$$

$$-4 = 0 + 0 + 0 + D(-2)$$

$$\Rightarrow -2D = -4$$

$$D = \frac{-4}{-2}$$

$$\boxed{D = 2}$$

Putting $x-1=0 \Rightarrow x=1$ in equation (ii), we get

$$4(1)^3 = A(1+1)^3 + B(0)(1+1)^2 + C(0)(1+1) + D(0)$$

$$4 = 8A + 0 + 0 + 0$$

$$\Rightarrow 8A = 4$$

$$A = \frac{4}{8}$$

$$\boxed{A = \frac{1}{2}}$$

Expanding equation (ii), we get

$$4x^3 = A(x^3 + 1 + 3x^2 + 3x) + B(x-1)(x^2 + 2x + 1) + C(x^2 - 1) + D(x-1)$$

$$4x^3 = A(x^3 + 1 + 3x^2 + 3x) + B(x^3 + x^2 - x - 1) + C(x^2 - 1) + D(x-1)$$

$$4x^3 + 0x^2 + 0x + 0 = (A+B)x^3 + (3A+B+C)x^2 + (3A-B+D)x + A-B-C-D \quad (iii)$$

Comparing coefficients of x^3 from equation (iii), we get

$$4 = A + B$$

$$\Rightarrow B = 4 - A$$

$$B = 4 - \frac{1}{2} \quad \because A = \frac{1}{2}$$

$$B = \frac{8-1}{2}$$

$$B = \frac{7}{2}$$

Comparing Coefficients of x^2 from equation (iii), we get

$$0 = 3A + B + C$$

$$\Rightarrow C = -3A - B$$

$$C = -3\left(\frac{1}{2}\right) - \frac{7}{2}$$

$$C = \frac{-3}{2} + \frac{-7}{2}$$

$$C = \frac{-10}{2}$$

$$C = -5$$

Putting the values of A , B , C and D in equation (i)

$$\frac{4x^3}{(x-1)(x+1)^3} = \frac{\frac{1}{2}}{x-1} + \frac{\frac{7}{2}}{x+1} + \frac{-5}{(x+1)^2} + \frac{2}{(x+1)^2}$$

$$\frac{4x^3}{(x-1)(x+1)^3} = \frac{1}{2(x-1)} + \frac{7}{2(x+1)} - \frac{5}{(x+1)^2} + \frac{2}{(x+1)^3}$$

Q:11 Resolve, $\frac{2x+1}{(x+3)(x-1)(x+2)^2}$ into Partial Fraction .

Solution:

$$\text{Let } \frac{2x+1}{(x+3)(x-1)(x+2)^2} = \frac{A}{x+3} + \frac{B}{x-1} + \frac{C}{x+2} + \frac{D}{(x+2)^2} \quad (i)$$

Multiplying by $(x+3)(x-1)(x+2)^2$ on both sides of equation (i)

$$2x+1 = A(x-1)(x+2)^2 + B(x+3)(x+2)^2 + C(x+3)(x-1)(x+2) + D(x+3)(x-1) \quad (ii)$$

Putting $x+3=0 \Rightarrow x=-3$ in equation (ii)

$$2(-3)+1 = A(-3-1)(-3+2)^2 + B(0)(-3+2)^2 + C(0)(-3-1)(-3+2) + D(0)(-3-1)$$

$$-6+1 = A(-4)(1) + 0 + 0 + 0$$

$$-5 = -4A$$

$$\Rightarrow A = \frac{-5}{-4}$$

$$\boxed{A = \frac{5}{4}}$$

Putting $x-1=0 \Rightarrow x=1$ in equation (ii)

$$2(1)+1=A(0)(1+2)^2+B(1+3)(1+2)^2+C(1+3)(0)(1+2)+D(1+3)(0)$$

$$3=0+B(4)(9)+0+0$$

$$3=36B$$

$$\Rightarrow B = \frac{3}{36}$$

$$\Rightarrow \boxed{B = \frac{1}{12}}$$

Putting $x+2=0 \Rightarrow x=-2$ in equation (ii), we get

$$2(-2)+1=A(-2-1)(0)^2+B(-2+3)(0)^2+C(-2+3)(-2-1)(0)+D(-2+3)(-2-1)$$

$$-3=0+0+0+D(1)(-3)$$

$$\Rightarrow -3D = -3$$

$$D = \frac{-3}{-3}$$

$$\boxed{D = 1}$$

Expanding equation (ii), we get

$$2x+1=A(x-1)(x^2+4x+4)+B(x+3)(x^2+4x+4)+C(x+3)(x^2+x-2)+D(x^2+2x-3)$$

$$2x+1=A(x^3+3x^2-4)+B(x^3+7x^2+16x+12)+C(x^3+4x^2+x-6)+D(x^2+2x-3)$$

$$0x^3+0x^2+2x+1=(A+B+C)x^3+(3A+7B+4C+D)x^2+(16B+C+2D)x-4A+12B-6C-3D$$

(iii)

Comparing Coefficients of x^3

$$0=A+B+C$$

$$\Rightarrow C=-A-B$$

$$C=-\frac{5}{4}-\frac{1}{12}$$

$$\therefore A=\frac{5}{4}, B=\frac{1}{2}$$

$$C=\frac{-15-1}{12}$$

$$C=\frac{-16}{12}$$

$$C = \frac{-4}{3}$$

Putting the values of A, B, C and D in equation (i), we get

$$\frac{2x+1}{(x+3)(x-1)(x+2)^2} = \frac{\frac{5}{4}}{x+3} + \frac{\frac{1}{12}}{x-1} + \frac{\frac{-4}{3}}{x+2} + \frac{1}{(x+2)^2}$$

$$\frac{2x+1}{(x+3)(x-1)(x+2)^2} = \frac{5}{4(x+3)} + \frac{1}{12(x-1)} - \frac{4}{3(x+2)} + \frac{1}{(x+2)^2}$$

Q:12 Resolve $\frac{2x^4}{(x-3)(x+2)^2}$ into Partial Fraction.

Solution:

It is improper rational Fraction, so first we transform it into mixed form.

$$\begin{array}{r} \frac{2x^4}{(x-3)(x^2+4x+4)} = \frac{2x^4}{x^3+x^2-8x-12} \\ \begin{array}{r} 2x-2 \\ x^3+x^2-8x-12 \overline{) 2x^4} \\ \underline{-2x^4 \pm 2x^3 \mp 16x^2 \mp 24x} \\ -2x^3 + 16x^2 + 24x \\ \underline{\mp 2x^3 \mp 2x^2 \pm 16x \pm 24} \\ 18x^2 + 8x - 24 \end{array} \end{array}$$

So

$$\frac{2x^4}{(x-3)(x+2)^2} = 2x-2 + \frac{18x^2+8x-24}{(x-3)(x+2)^2} \quad (i)$$

$$\text{Let, } \frac{18x^2+8x-24}{(x-3)(x+2)^2} = \frac{A}{x-3} + \frac{B}{x+2} + \frac{C}{(x+2)^2} \quad (ii)$$

Multiplying by $(x-3)(x+2)^2$ on both sides of equation (ii)

$$18x^2+8x-24 = A(x+2)^2 + B(x-3)(x+2) + C(x-3) \quad (iii)$$

Putting $x-3=0 \Rightarrow x=3$ in equation (iii)

$$18(3)^2+8(3)-24 = A(3+2)^2 + B(0)(3+2) + C(0)$$

$$162 = A(25) + 0 + 0$$

$$25A = 162$$

$$\Rightarrow A = \frac{162}{25}$$

Putting $x + 2 = 0 \Rightarrow x = -2$ in equation (iii)

$$18(-2)^2 + 8(-2) - 24 = A(0)^2 + B(-2-3)(0) + C(-2-3)$$

$$18(4) - 16 - 24 = 0 + 0 - 5C$$

$$32 = -5C$$

$$\Rightarrow C = \frac{-32}{5}$$

Expanding equation (iii), we get.

$$18x^2 + 8x - 24 = A(x^2 + 4x + 4) + B(x^2 - x - 6) + C(x - 3)$$

$$18x^2 + 8x - 24 = (A + B)x^2 + (4A - B + C)x + 4A - 6B - 3C \quad (\text{iv})$$

Comparing Coefficient of x^2 from equation (iv)

$$18 = A + B$$

$$\Rightarrow B = 18 - A$$

$$B = 18 - \frac{162}{25}$$

$$B = \frac{450 - 162}{25}$$

$$B = \frac{288}{25}$$

Putting the values of A , B and C in equation (ii), we get

$$\frac{18x^2 + 8x - 24}{(x-3)(x+2)^2} = \frac{\frac{162}{25}}{x-3} + \frac{\frac{288}{25}}{x+2} + \frac{\frac{-32}{5}}{(x+2)^2}$$

Now by using equation (i), partial fractions are:

$$\frac{2x^4}{(x-3)(x+2)^2} = 2x - 2 + \frac{162}{25(x-3)} + \frac{288}{25(x+2)} - \frac{32}{5(x+2)^2}$$